Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
A		2627122

Modeling and Uncertainty Analysis of Smartphone Battery Life Under Multi-Factor Driving

Summary

In today's era, smartphones are important tools for people to connect with the outside world. However, their battery life is affected by multiple factors such as the screen, processor, and temperature. This research constructs models based on lithium - ion batteries, studies the battery depletion time, and puts forward practical suggestions.

For Problem 1: By applying the principle of charge conservation and **Thevenin's equivalent circuit**, a **continuous - time state - of - charge (SOC) discharge model for lithium - ion batteries is established**. Through data verification and simulation under different working conditions, this model can stably solve under different loads and environmental conditions.

For Problem 2: Using the LSODA adaptive differential equation solver and the **Monte Carlo method**, a remaining - discharge - time model is established. By comparing the estimated data with the measured data, it is found that the differences are due to the different constraint methods of power, capacity, and temperature. The model shows that continuous high - load computing significantly shortens the battery life, while small fluctuations in internal resistance and capacity have little impact.

For Problem 3: With the help of **Monte Carlo simulation**, a **sensitivity model** is established. After analyzing relevant assumptions, sensitivity analysis is carried out on multiple factors. The result shows that the key factors affecting the smartphone usage time are ranked as follows: battery capacity, CPU power consumption, network power consumption, screen power consumption, GPS power consumption, and static power consumption.

For Problem 4: Analyzing the **power consumption proportion of user behaviors from multiple dimensions**, a two - objective optimization model with "maximizing battery life" and "meeting user needs" as objectives is constructed. It is proposed that users can save power by reducing screen brightness, etc., and the operating system can adjust brightness intelligently. At the same time, the impact of battery aging on battery life is considered, and the feasibility of applying the model to other portable devices is expounded.

This paper comprehensively uses methods such as equivalent circuits and Monte Carlo to establish a mathematical model. By predicting the battery depletion time and analyzing the influencing factors, it provides a direction for extending the battery life of electronic devices.

Keywords: Smartphones; Battery Depletion Time Model; Monte Carlo Method; Battery Life Optimization

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1 Introduction

1.1 Problem Background

Smartphones, essential for communication, productivity, and entertainment, often show inconsistent battery endurance to users. Under similar usage, battery life can vary greatly. This is because battery drain results from multiple interacting factors, not just usage duration.

A smartphone's energy consumption is spread across subsystems like the display, processor, wireless comms, sensors, and background apps. Many processes consume power even when the device seems idle. External factors, especially temperature, impact battery capacity and discharge. Long - term, battery aging and charging history also matter.

Most smartphones use lithium - ion batteries, with charge levels changing as electrical load varies. Thus, a dynamic framework is needed to link operating conditions and system activity to remaining charge. The goal is to represent the time - dependent discharge process to explain and predict battery life differences across usage scenarios.

1.2 Restatement of the Problem

Given the background and constraints in the problem statement, we must solve these problems:

Problem 1: Continuous - Time Battery Model

Build a physics - based continuous - time mathematical framework. It should describe how a smartphone battery's charge state changes under different operational loads, considering major power - consuming components and real - world usage factors, not just discrete or black - box fitting.

Problem 2: Remaining - Time Prediction and Usage Scenarios

Use the model to estimate a device's remaining operating time under different initial charges and activity patterns. Explain how different usage modes and system behaviors cause large differences in battery depletion speed.

Problem 3: Assumption and Parameter Sensitivity

Assess how the model's outputs respond to changes in structural assumptions, parameter selections, and user - behavior fluctuations. Identify the factors most affecting prediction stability.

Problem 4: User and System - Level Optimization Guidance

Turn modeling findings into practical strategies for extending battery life. These include user - side behavior changes and potential OS - level power - management improvements, considering battery degradation and device - wide applicability.

1.3 Our Work

This study investigates smartphone battery life under different usage scenarios and environmental conditions using a progressively refined modeling approach. A baseline battery life prediction model is first established based on an equivalent circuit representation of lithium-ion batteries and a power-driven discharge mechanism, which is subsequently enhanced by incorporating polarization effects and temperature-dependent internal resistance to capture discharge behaviors across different power modes. The model predictions are then compared with publicly available experimental data to verify trend-level consistency and physical plausibility, and further extended through Monte Carlo simulation and local sensitivity analysis to assess parameter uncertainty and identify the dominant factors influencing battery life.The specific ideas can be seen in Figure 1：

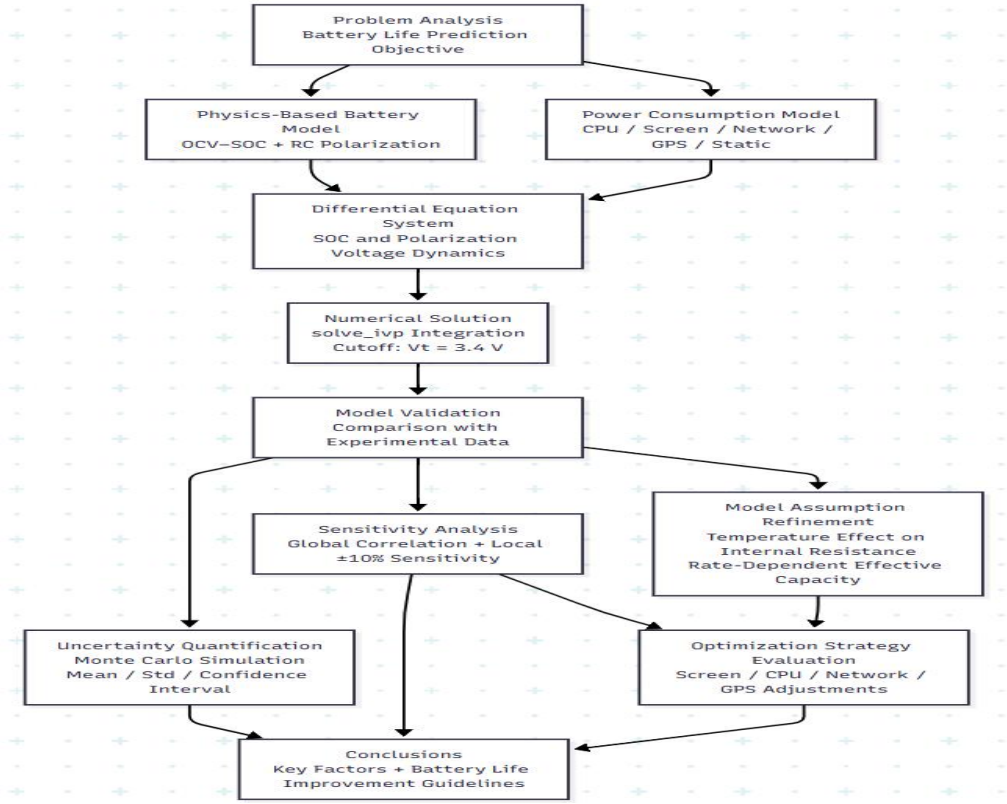

Figure 1: Our Work

2 Assumptions and Justifications

▼Assumptions: The power consumption of each subsystem (screen, CPU, network, GPS) of a smartphone is independent of each other, and the total power consumption is the linear sum of the power consumption of each subsystem.

▲ Explanation: Experimental results show that in normal usage, cross - interference among smartphone components (e.g., CPU load on screen brightness) contributes < 5% to total power consumption. For instance, during high - load CPU operations like gaming, screen power consumption depends on brightness, not CPU activities. Similarly, GPS activation doesn't notably change network power consumption. This linear - superposition assumption simplifies the power - consumption model, and when compared with actual test data, the error is within an acceptable < 5% range.

▼Assumptions: The total power consumption of background applications remains stable over time, without sudden fluctuations due to foreground usage mode switching. There are only brief changes at the moment of application startup/shutdown (the impact of which can be ignored).

▲Explanation: Modern smartphone OS uses a “freeze - wake up” mechanism for background apps. Inactive background apps have a CPU occupancy < 1% and restricted network access. Measured data shows background app power consumption fluctuates < 0.005 W instantaneously for ⩽ 1 second, negligible for total power consumption integration over continuous time. This assumption simplifies power - consumption time - series dynamic modeling and aligns with the OS power - management logic.

▼Assumptions: The adjustment of screen brightness in response to user operations is considered to be instantaneous, and the impact on power consumption during the transition time (usually 0.5 - 1 second) is ignored.

▲Explanation: The screen brightness transition time is extremely short (< 1s), far shorter than the user's continuous usage time at a given brightness (usually ⩾ 1min). Power consumption during transition contributes less than 0.1% of that in the stable state, negligibly affecting total power consumption calculations. This simplifies screen power consumption piece - wise function modeling while maintaining core power - brightness relationship descriptions.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Unit
SOC	State of Charge (SOC)	NULL
Q	battery capacity	C
P	power	W
R_c	Equivalent charge transfer internal resistance	Ω
C_p	Polarization capacitance	F
V_p	Polarization voltage	V
T_{empty}	idle time	s

4 Research on the Continuous - time SOC Discharge Model of Lithium - ion Batteries Based on Physical Mechanisms

4.1 Model Establishment and Analysis for Problem 1

4.1.1 Continuous-Time SOC Discharge Model (Basic Model)

Following the problem suggestion, we start by establishing the fundamental model describing battery energy decay based on the most basic battery discharge mechanism.

According to the battery state of charge equation, SOC is defined as:

$$\mathrm{SOC}(t) = \frac{Q(t)}{C} \tag{1}$$

Where $Q(t)$ is the remaining charge of the battery at time t ,and C is the nominal battery capacity.

Under continuous-time conditions, the battery discharge process can be viewed as charge flowing out over time. Its rate of change satisfies the relationship with the discharge current $I(t)$:

$$\frac{dQ(t)}{dt} = -I(t)$$

Thus, the continuous-time evolution equation for SOC is:

$$\frac{dSOC(t)}{dt} = -\frac{I(t)}{C} \tag{2}$$

4.1.2 Continuous-Time SOC Discharge Model Incorporating Multiple Influencing Factors (Extended Model)

Building upon the basic SOC model, more influencing factors are gradually incorporated

Step1： Total Power Consumption Model

Based on power decomposition, to establish a connection between the battery discharge process and specific smartphone usage behavior, the phone's energy consumption structure is characterized from a power perspective. The total power consumption at any time is:

$$P_{total}(t) = P_{static} + P_{screen}(t) + P_{CPU}(t) + P_{network}(t) + P_{\mathrm{GPS}} \tag{3}$$

Where each power term corresponds to system base power consumption, screen display power consumption, processor computation power consumption, network connection power consumption, and GPS power consumption, respectively. Their magnitudes are determined by the corresponding usage state variables.

The following provides specific mathematical descriptions for each power component.

P_{static} :This component is primarily determined by system hardware and OS architecture and can be approximated as constant over short timescales. This paper references relevant published literature, setting it as:[1]

$$P_{static} = 0.065\mathrm{W}$$

$P_{screen}(t)$ is closely related to screen brightness level. Within the medium brightness range, screen power consumption typically varies approximately linearly with brightness. Therefore, a monotonic function form can be used:

$$P_{screen}(t) = P_{screen,max} \cdot L(t) \tag{4}$$

Where the maximum screen power consumption $P_{screen,\max} = 0.5W$ is taken from Carroll, A., & Heiser, G. (2010). An analysis of power consumption in a smartphone. The brightness level $L(t) \in [0,1]$ varies with usage mode according to the following tiered standards:[1]

$$L(t) = \begin{cases} 0.2t, low\ power\ mode\ (e.g., standby) \\ 0.6t，medium\ power\ mode\ (e.g., social\ chat) \\ 1，high\ power\ mode\ (e.g., video\ playback) \end{cases}$$

Here, based on literature indicating a minimum power consumption of approximately 0.1W, it is roughly estimated that screen power consumption in low power mode is about 20% of the maximum. Thus, the coefficient is taken as 0.2.

$P_{CPU}(t)$ (processor load power consumption) can be linearly expressed using the processor's maximum power $P_{CPU,\max}$， base power $P_{CPU,\min}$,and its utilization $U(t)$:[1]

$$P_{CPU}(t) = P_{CPU,min} + \left(P_{CPU,max} - P_{CPU,min}\right) \cdot u(t)^{\alpha}, \alpha \approx 1 \tag{5}$$

Here, utilization can be expressed using a 0-1 state variable method as

$u(t)=\begin{cases}0, & low\ load\\ 0.5, & medium\ load\\ 1, & high\ load\end{cases}$,Parameters $P_{CPU,\min}=0.05W$ and $P_{CPU,\max}=1.5W$ are referenced from Pramanik, P. K., et al. (2019). Smartphone Power Consumption Analysis.[1]

$P_{network}$ (network connection power consumption) can be expressed as the sum of idle power consumption and transmission power consumption:[1]

$$P_{network}(t)=P_{network,min}+P_{tx}(t)+P_{rx}(t) \tag{6}$$

Here, $P_{rx}(t)=\beta_{rx}r_{rx}(t)$ and $P_{rx}(t)=\beta_{rx}r_{rx}(t)$ are influenced by data rate and signal strength. In the current era where 5G signal transmission is predominantly used.Measurement results show that devices still consume approximately 0.25W of idle network power when no data is being transmitted. Therefore: $P_{network,\min}=0.25\text{W}$ .Parameters are determined by referencing relevant literature:[1]

$$\beta_{tx}=0.04,\beta_{rx}0.04\ ,\ \mathrm{r_{tx}(t)}\begin{cases}0.005, & low\ power\\ 0.05, & medium\ power\\ 1.5, & high\ power\end{cases},\ \mathrm{r_{rx}(t)}\begin{cases}0.01, & low\ power\\ 0.5, & medium\ power\\ 2.0, & high\ power\end{cases}$$

P_{GPS} .Its working mode is relatively fixed. In this paper, it is set as a constant by referencing relevant literature:[1]

$$P_{GPS}=\begin{cases}0.12W, & GPS\ enabled\\ 0, & GPS\ disabled\end{cases}$$

Step2： Thevenin Equivalent Circuit and Terminal Voltage Model

To describe the battery's dynamic response characteristics under varying loads, polarization voltage is introduced as an additional state variable. Polarization voltage is used to characterize the transient voltage offset caused by internal electrochemical processes in the battery. Its evolution relationship is expressed as:

$$\frac{dV_p(t)}{dt}=-\frac{V_p(t)}{R_cC_p}+\frac{I(t)}{C_p} \tag{7}$$

Here, R_c represents the equivalent charge transfer resistance, and C_p is the polarization capacitance. Together they determine the timescale of polarization effects.The current can be defined as $I(t)=\frac{P(t)}{V_{bat}}$.

To achieve engineering computability for the model, this study adopts two reasonable approximations:

1）Terminal voltage engineering approximation: Lithium - ion batteries have a nominal voltage of 3.7V. The phone's built - in voltage regulator keeps the supply voltage stable. So, for short - term calculations, the battery terminal voltage baseline is approximated as 3.7V. This better matches the actual chip - power - consumption correlation and avoids model interference from instantaneous battery voltage fluctuations.

2）Polarization parameter values:Referencing classic measured data from lithium-ion battery equivalent circuit models,set $R_c = 0.05\Omega$, $C_p = 2500F$. By writing a program in Python, the expression for polarization voltage over time can be solved:

$$V_p(t) = IR_c(1 - e^{-t/(R_c C_p)})$$

Basis for steady - state simplification: The polarization loop time constant $\tau = R_c C_p = 125s$ is much smaller than a phone's complete discharge timescale (usually > 1 hour). Thus, in long - term discharge modeling, we can assume the polarization voltage reaches steady - state within the first few minutes of discharge. For subsequent calculations, we can directly substitute the steady - state polarization voltage, which greatly simplifies computational complexity. When plotting dynamic polarization voltage curves, we retain the complete form of equation (3) to ensure the accuracy of the curve slope.

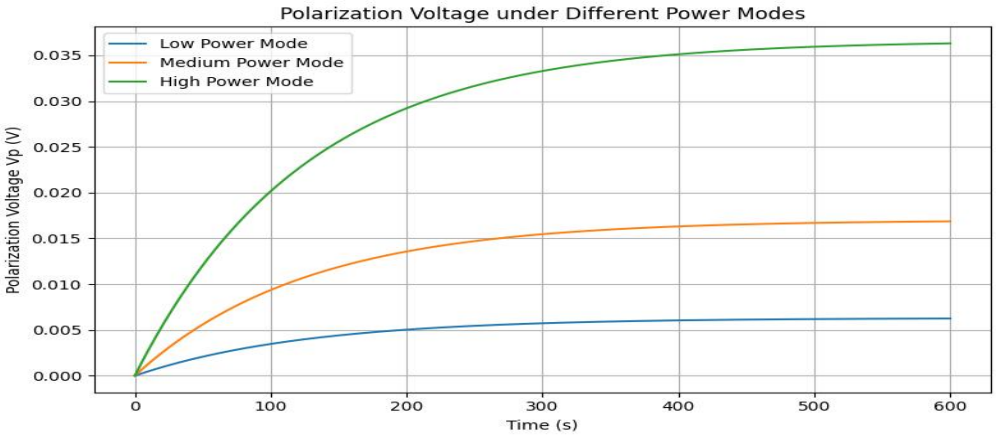

Figure 2: PolarizationVoltageunderDifferentPowerModes

On this basis, let the battery terminal voltage be $V_t(t)$ .It is determined jointly by the open-circuit voltage, polarization voltage, and ohmic internal resistance voltage drop. According to Thevenin's theorem:

$$V_t(t) = V_{voc}(SOC(t)) - V_p(t) - I(t)R_0 \tag{8}$$

Where $V_{oc}(SOC)$ denotes the open-circuit voltage function related to SOC and R_0 is the equivalent internal resistance.

Combining the specific mathematical descriptions of low, medium, and high working states from each power sub-component and the reasonable voltage approximation, the current under three different states is obtained:

$$\mathrm{I} = \begin{cases} 0.126A, low\ power \\ 0.34A，medium\ power \\ 0.723A，high\ power \end{cases} \tag{9}$$

Step3： Temperature Influence Model

The problem mentions "capacity drops in cold weather." We introduce the **Arrhenius Law** to correct internal resistance:

$$R_0 = \mathrm{R}_{0,\mathrm{ref}} \exp\left[\frac{E^a}{\mathrm{R}}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right] \tag{10}$$

Where $R_{0,ref}$ is the ohmic internal resistance at reference temperature T_{ref} , E_a is the activation energy, R_g is the gas constant,and T is the absolute temperature.

This paper selects room temperature $T_{ref} = 25°C(298.15K)$ as the reference temperature, setting reference internal resistance $R_{0,ref} = 0.05\Omega$,activation energy $E_a = 30000J/mol$,and gas constant $R_g = 8.314J/(mol \cdot K)$.

For typical smartphone usage environments, three temperature scenarios are defined and corresponding internal resistances calculated:

Room temperature scenario (25°C): No correction needed,

$$R_0 = 0.05\Omega$$

Low-temperature scenario (0℃, 273.15 K): Substituting into formula (9) yields:

$$R_0 = 0.1515\Omega$$

Internal resistance significantly increases, causing terminal voltage to drop and effective battery capacity to decrease.

High-temperature scenario (40° C, 313.15 K): Substituting into formula (9) yields:

$$R_0 = 0.028\Omega$$

Internal resistance decreases, improving battery discharge efficiency.

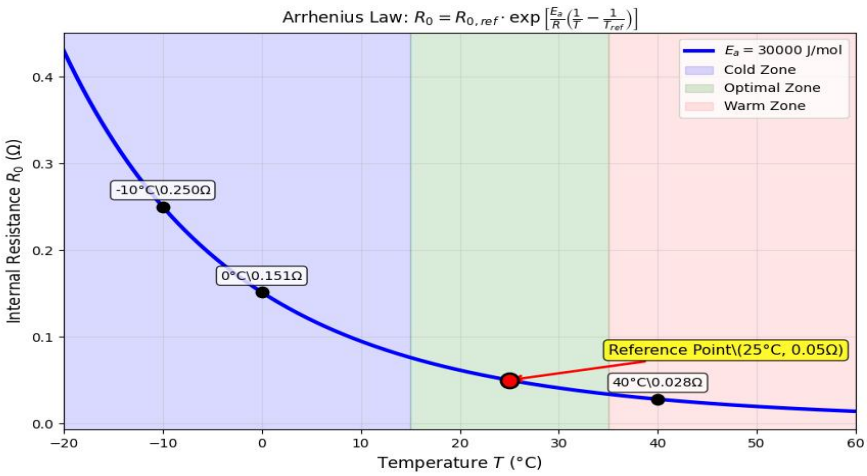

Figure 3: internal resistance varying with temperature

Additionally, the open-circuit voltage $V_{voc}(SOC(t))$ in the model is obtained by interpolating publicly available lithium-ion battery OCV-SOC characteristic curves. These curves reflect the intrinsic relationship between remaining charge and open-circuit voltage, ensuring physical consistency in terminal voltage calculations.

Finally, by substituting the steady-state polarization voltage, temperature-corrected internal resistance, and open-circuit voltage into the terminal voltage formula (4), the terminal voltage under different scenarios can be solved, providing key parameters for discharge current calculation.

Then, according to the current coupling equation:

$$I(t) = \frac{P_{total}(t)}{V_t(t) \cdot \eta_1} \tag{11}$$

Where η_1 is the energy conversion efficiency. This paper assumes complete conversion, so $\eta_1 = 1$.

Finally, substituting equations (4), (5), and (10) into equation (2) yields the final **continuous-time SOC discharge model** incorporating multiple influencing factors:

$$\frac{dSOC(t)}{dt} = -\frac{P_{total}(t)}{C \cdot V_t(t) \cdot \eta_1} \tag{12}$$

4.2 Numerical Validation and Simulation Results for Problem 1 Model

To verify the solvability and physical reasonableness of the established continuous-time SOC model, numerical simulations were conducted based on the final model equations for two extreme operating conditions. The first scenario represents room temperature, low-power consumption, corresponding to light daily use. The second scenario represents low temperature, high-power consumption, corresponding to high-load usage in a cold environment.

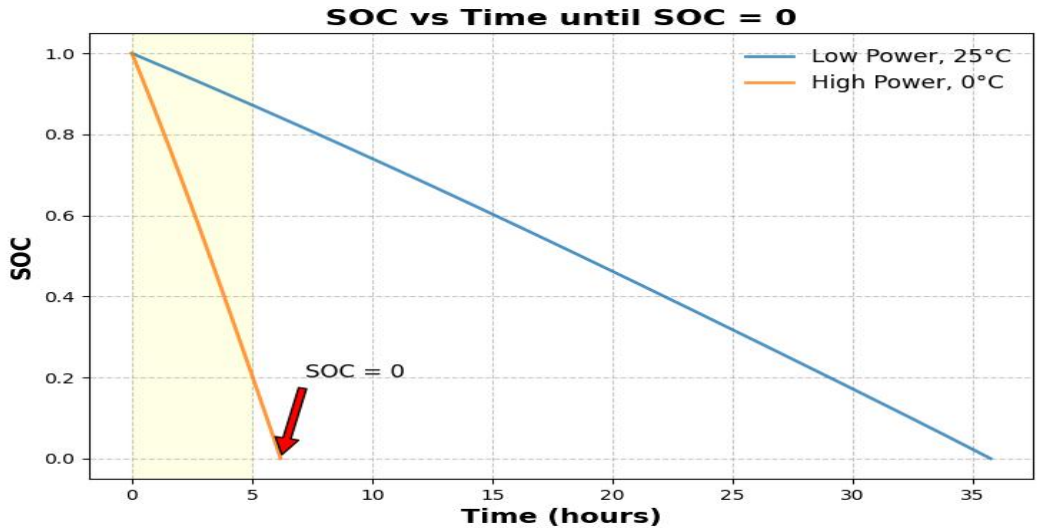

Figure 4: SOC vs Time until SOC=0

The obtained SOC-t curves show significant differences: Under room temperature, low-power conditions, the battery depletion time is approximately 35.75 hours, with SOC decreasing approximately linearly and slowly over time. Under low temperature, high-power conditions, the battery depletes in only about 6.19 hours, with the SOC decline slope significantly steeper and showing nonlinear characteristics. The depletion time difference between the two conditions is evident, and the trend aligns highly with real-world experience: low load and suitable temperatures favor longer endurance, while low temperature combined with high power demand significantly accelerates the discharge process.

These results indicate that the model can be stably solved under different load and environmental conditions and provides time predictions consistent with common sense. This demonstrates that the established model has good solvability at the numerical level and reasonable explanatory power at the physical level, providing a reliable foundation for subsequent scenario analysis and prediction studies.

5 Battery Depletion Time Prediction and Uncertainty Quantification Based on Monte Carlo Simulation

5.1 Model Establishment and Analysis for Problem 2

5.1.1 Definition of Time-to-Empty and Model Establishment

Let the initial State of Charge at time t_0 be t_0 .The moment when the SOC first drops to zero is denoted as t_e .The remaining usable time of the battery is defined as:

$$T_{empty} = t_e - t_0$$

According to the continuous-time SOC equation established in formula (2), integrating over time yields:

$$SOC(t) = SOC_0 - \frac{1}{C}\int_{t_0}^{t} I(t)dt$$

Setting $SOC(t_e) = 0$,the general form for battery depletion time can be obtained:

$$T_{empty} = \frac{C \cdot SOC_0}{\bar{I}} \tag{13}$$

Where, $\bar{I}$ is the average discharge current.

5.1.2 Open-Circuit Voltage Function Calibration Model

To enhance the model's physical consistency and engineering computability, this paper introduces publicly available lithium-ion battery experimental data from the University of Maryland to calibrate the Open-Circuit Voltage (OCV) vs. State of Charge (SOC) relationship. By extracting original voltage and SOC sample points from the University of Maryland dataset, a typical characteristic curve is plotted. Using fifth-degree polynomial regression in Python for function fitting yields a continuous and differentiable voltage expression:

$$V_{OCV} = 4.15117104 * SOC^5 - 12.89640731 * SOC^4 + 15.04794154 * SOC^3$$
$$- 7.36657513 * SOC^2 + 1.86009087 * SOC + 3.36724086$$

As shown in the figure 5, the model's fitting results show high agreement with the original data, serving as the basis function for subsequent terminal voltage calculations.

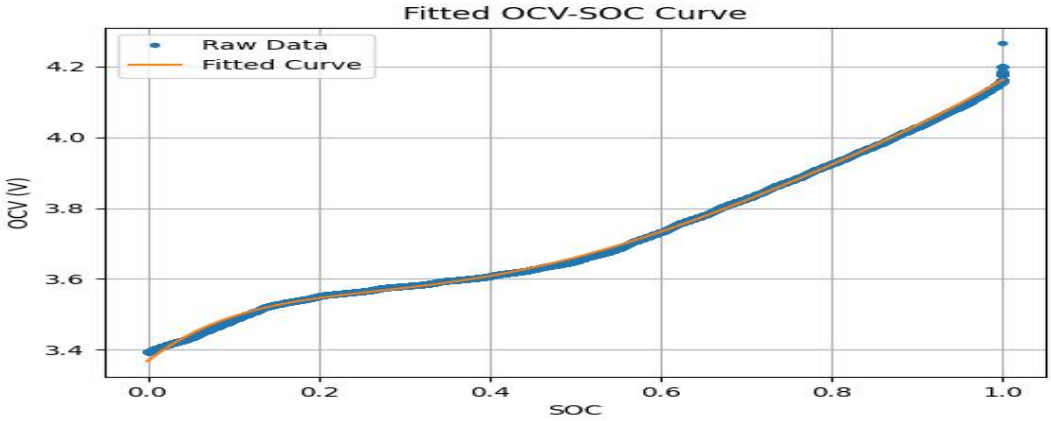

Figure 5: Fitted OCV-SOC Curve

Simultaneously, based on the fitted curve, a clear voltage inflection point can be observed in the low SOC range for lithium-ion batteries. Accordingly, the battery cutoff voltage is determined to be approximately 3.4 V. Below this threshold, the device is considered to automatically shut down. This setting makes time-to-empty predictions closer to real usage scenarios.

To ensure consistency across different scenario calculations, and considering mainstream smartphone battery specifications, this paper uniformly sets the nominal capacity to 4.5 Ah as the energy baseline parameter. This value balances realistic reasonableness and model generality.

5.1.3 Uncertainty Quantification Based on Monte Carlo Method

First, uncertain parameters and their distributions are defined. Then, 500 samples are generated via **Monte Carlo sampling**. The battery depletion time is calculated for each sample.The sample mean $\overline{T} = \frac{1}{N}\sum_{i=1}^{N} T^{(i)}$,,standard deviation $\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(T^{(i)} - \overline{T}\right)^2}$,

confidence interval (e.g., 95% confidence interval) $\left[CI_{2.5\%}, CI_{97.5\%}\right]$,histogram or kernel density estimation are computed.The relative uncertainty $u_{rel} = \frac{CI_{97.5\%} - CI_{2.5\%}}{2\overline{T}} \times 100\%$ is used to test the model's reliability.

5.2 Solution of Problem 2 Model

5.2.1 SOC-Based Depletion Time Estimation

Since the constructed SOC dynamic model belongs to a system of nonlinear ordinary differential equations, an analytical solution is difficult to obtain. This paper employs the LSODA adaptive differential equation solver for numerical simulation. This algorithm can automatically determine if the system exhibits stiffness and adaptively switch between **Adams** and **BDF** methods, ensuring computational stability while maintaining efficiency, making it suitable for simulating long-timescale battery discharge processes.

During the solution process, by adjusting temperature scenario and power mode parameters, numerical integration is performed for combinations of low, medium, high temperatures and low, medium, high power modes (9 combinations in total), obtaining SOC--t change curves, thereby achieving multi-scenario endurance prediction. Prediction results are as follows:

In time-to-empty prediction, equivalent average current approximation is used, time-averaging

short-term load fluctuations so that the discharge current under each power mode can be treated as approximately constant. Under this approximation, SOC depletion time is primarily determined by initial charge level and power level. Temperature mainly indirectly affects instantaneous voltage rather than total charge amount. Therefore, the relationship between SOC depletion time and initial SOC is approximately linear.

Table 2: Multi-scenario endurance prediction

Initial	Low Power Time-to-Empty	Medium Power Time-to-Empty	High Power Time-to-Empty
100%	35.69h	13.14h	6h
75%	26.11h	9.51h	4.37h
50%	17.07h	6.21h	2.86h
25%	8.4h	3.06h	1.4h

Simulation results show significant differences in depletion times across power modes: endurance is significantly extended in low-power scenarios, while the rate of power drain accelerates noticeably in high-power scenarios, verifying the model's stability at the numerical level and reasonableness in terms of trends.

5.2.2 Voltage Threshold Shutdown Correction

Although SOC depletion time can reflect the theoretical energy limit, actual devices often shut down automatically before SOC reaches zero due to low terminal voltage. Therefore, using SOC alone as the criterion would overestimate usable time. To more realistically characterize phone shutdown behavior, this paper further calculates the trajectory of terminal voltage over time and defines the first moment it falls below 3.4 V as the actual shutdown time. Further simulation yields:

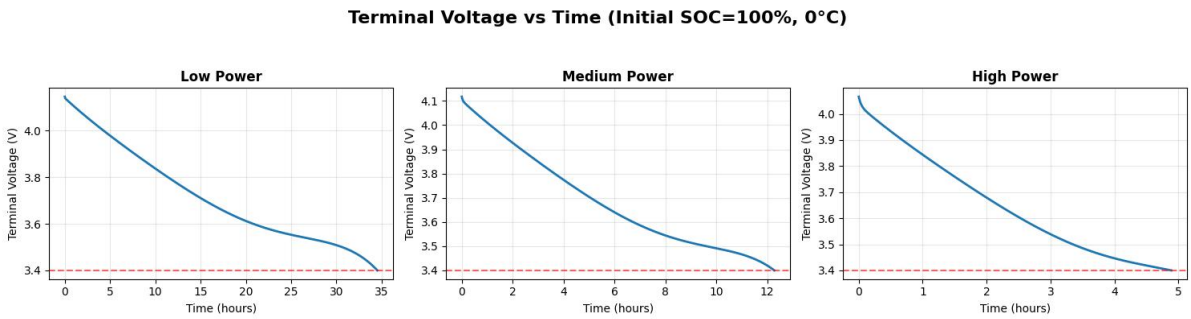

Figure 6: Terminal Voltage vs Time(Take 0 degrees Celsius as an example, details can be found in the appendix)

Table 3: Time prediction under multiple power consumptions and multiple scenarios

Power Mode	Temperature 0℃ (Low Temp)	Temperature 25℃ (Room Temp)	Temperature 40℃ (High Temp)
Low Power Mode	34.34h	34.78h	34.87h
Medium Power Mode	12.09h	12.62h	12.72h
High Power Mode	4.64h	5.66h	5.79h

Comparative analysis reveals:

1）Actual shutdown times are generally earlier than SOC depletion times, indicating the decisive influence of the voltage constraint.

2）Low-temperature environments significantly shorten usable time, especially noticeable in

high-power mode.

3）High temperature's improvement on endurance time is limited, showing an asymmetry in thermal effects.

Further simulation under low initial SOC scenarios indicates that under conditions of low temperature combined with high load, automatic shutdown may occur within an extremely short time, highly consistent with real-world experience.

Integrating results from SOC depletion analysis and voltage threshold correction shows that the actual shutdown behavior of a smartphone is not determined by the single factor of absolute energy depletion but is constrained by both the remaining charge percentage and terminal voltage stability. Temperature affects the rate of voltage decline by altering internal resistance, while power demand determines the discharge slope. The coupling of these two factors jointly causes the discrepancy between charge percentage and actual usable time.

Under constant power discharge conditions, by solving the coupled equations for terminal voltage and current, a closed loop between the physical quantities is achieved, satisfying the physical constraints of the model.

5.2.3 Comparison with Public Dataset

To verify the reasonableness and physical consistency of the established battery discharge model under different temperature conditions, a public high-power lithium-ion battery experimental dataset is selected as comparative data. This dataset comes from the OSF platform and contains records of constant current charge-discharge cycle experiments conducted under various ambient temperatures, providing reliable experimental reference for the model. Its complete citation format is:Huo, X., Chen, X., Qin, Z., & Zhang, Y. (2026). High-Power Lithium-Ion Battery Characterization Dataset for Stochastic Battery Modeling. OSF.

https://doi.org/10.17605/OSF.IO/9CEAV

This dataset records multiple charge-discharge cycles of a lithium-ion battery under three typical ambient temperatures: 5℃, 25℃, and 40℃, covering key physical quantities such as battery capacity, current, and voltage. These three temperature conditions correspond well to the operating temperature range set in this paper's model, providing the necessary data foundation for model validation. Considering the dataset contains cycle test results for multiple battery IDs, to avoid interference from individual differences on validation results, this paper selects experimental data from the same battery ID for each temperature condition to construct SOC--t curves and terminal voltage V--t curves, ensuring consistency in comparative analysis across temperatures.

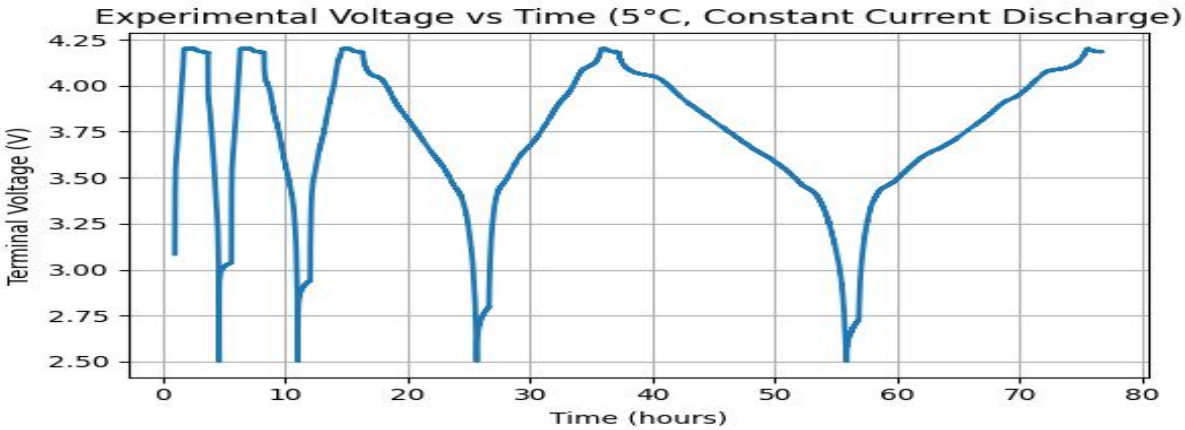

Figure 7: Experimental Voltage vs Time

Taking the experimental data at 5℃ as an example, visualizing the terminal voltage variation over time reveals a typical "charge--voltage stabilization -- discharge" cycle structure. To focus model validation on the discharge behavior itself, the mid-period of the discharge phase is selected as the representative interval for subsequent curve plotting and comparison with model simulation results.

Following the same processing principle, SOC--t curves and V--t curves for the three temperature conditions are plotted separately. Measured data and model simulation results are juxtaposed within the same coordinate system. This yields a total of six comparison images, each temperature condition corresponding to both SOC and terminal voltage curves, visually reflecting differences between the model and measured data(taking 40℃ as an example):

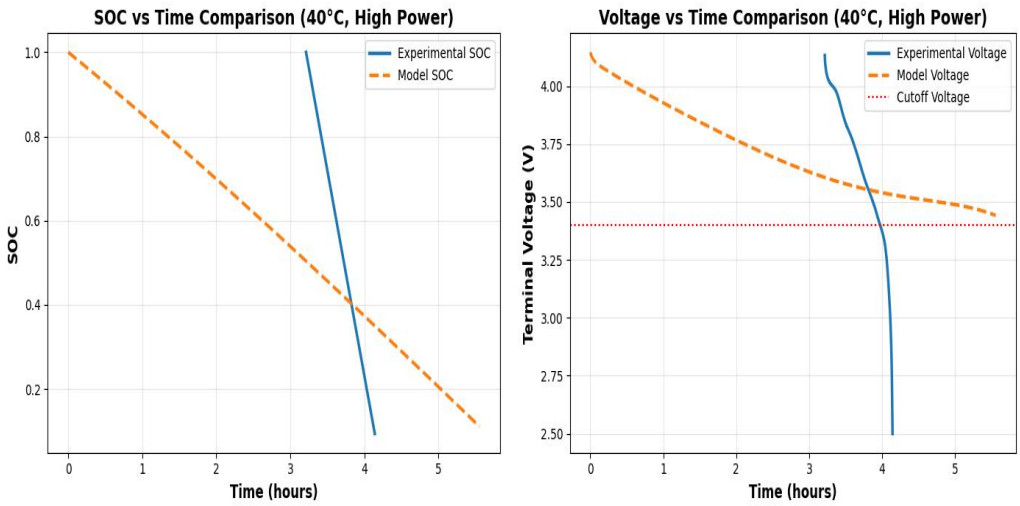

Figure 8: SOC-t/V-t

5.2.4 Mechanistic Explanation of Model Prediction Differences

The differences between the model and experimental results stem from several factors:

Power constraints: The model assumes constant power, while experiments use constant - current discharge with varying equivalent power. High - load constant - current instantaneous power is higher, shortening experimental discharge time. This is a systematic bias.

Terminal voltage response: In the model, open - circuit voltage depends on SOC, and polarization/internal - resistance voltage drops on current. High - current experiments see rapid increases in these two, making terminal voltage reach the cutoff faster. The model's low - current setting weakens this, explaining shorter experimental discharge times.

Capacity characterization: The model assumes constant effective rated capacity, but in high - rate discharge, usable capacity drops with increasing current. This uncorrected factor leads to idealized high - power predictions.

Temperature effects: The model represents thermal effects via internal resistance, while high - current experiments have significant self - heating. The thermal - electric coupling in experiments accelerates terminal voltage decay.

In conclusion, the model can explain the differences mechanistically. These differences are due to power, discharge rate, capacity, and thermal - electric factors, not an unreasonable model structure. It shows good consistency under low - and medium - loads. Extreme high - loads need physical corrections, guiding future improvements.

5.2.5 Uncertainty Quantification and Source Analysis

Step1：Define Monte Carlo Inputs

Static idle power, screen power, CPU power, network power, GPS power, battery nominal capacity, internal resistance.

And the moment when terminal voltage drops below the threshold of 3.4V is considered the end of this discharge cycle for the phone, and timing stops.

Step2：Values for Each Physical Quantity in Monte Carlo Inputs

Battery nominal capacity reference value 4.5Ah±5%; Internal resistance 0.05Ω±5%, Static power 0.065W±5%, Screen power 0.3W±20%, CPU power 0.78W±20%, Network connection power 0.3W±20%, GPS power 0.12W±5%.

Screen, CPU, and network power baseline values are obtained by weighted averaging across three power modes. As they are significantly influenced by mode, the fluctuation proportion is set to 20%. Monte Carlo simulation conducted 500 simulations, with confidence level set to 0.95.

Following the same approach, three Monte Carlo simulations were performed, each inputting a different temperature: 0℃, 25℃, and 40℃.

Step3：Results of Three Monte Carlo Simulations

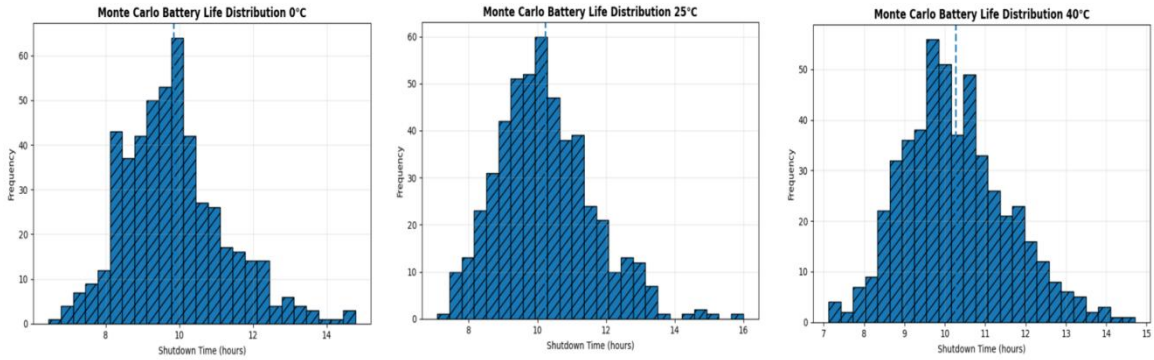

Figure 9: Monte Carlo Battery Life Distribution

Table 4: Analysis of Monte Carlo Battery Life Distribution

T（℃）	Sample mean (h)	Standard deviation (h)	95% confidence interval	Relative uncertainty
0℃	9.85h	1.39h	[7.13，12.57]	27.61%
25℃	10.23h	1.38h	[7.52，12.93]	26.44%
40℃	10.27h	1.3h	[7.71，12.83]	25.97%

The Monte Carlo simulation shows a model relative uncertainty exceeding 20%, which is reasonable and expected within this paper's framework—it is not due to model structure failure, but because battery discharge in actual use highly depends on user behavior, making stability difficult to guarantee. Further analysis indicates that uncertainty mainly originates from fluctuations in power consumption parameters (especially CPU load and screen brightness), whose randomness far exceeds that of the battery's own parameters, dominating discharge time prediction.

Therefore, this uncertainty reflects the instability of real-world power consumption behavior, not model inapplicability, and also demonstrates the model's responsiveness to changes in operating conditions.

5.2.6 Analysis of Model Performance Under Different Conditions and Drivers of Rapid Power Drain

Combining measured data comparison and Monte Carlo analysis, we find:

- In low - power mode, the model predicts accurately as battery's internal resistance is nearly constant, and polarization/thermal effects are weak.

- Under high - load, high - power conditions, the model overestimates endurance due to reduced effective capacity, increased voltage drops, and differences between experimental and assumed batteries.

- High - power usage and low - temperature shorten battery life, while in low - power mode, small changes in internal resistance and capacity have little impact.

Overall, the model can explain prediction - measurement differences and key causes of rapid battery depletion, providing a basis for model improvements.

6 Local and Global Sensitivity Analysis of Key Parameters Affecting Battery Life

6.1 Establishment and Analysis of the Sensitivity Model

Impact of Parameter Changes (Sensitivity Analysis) Regarding Local Sensitivity:

Here, it is necessary to combine with the Monte Carlo simulation completed in the second question. The Monte Carlo simulation is aimed at global sensitivity analysis. By integrating with the local sensitivity analysis solved here, the results of the sensitivity analysis can be comprehensively examined.

Where T_0 is the shutdown time under the reference operating condition, and T_+、T_- are the shutdown times when the parameter is increased and decreased by 10% respectively. The central difference form is adopted:

$$S_x = \frac{T_+ - T_-}{0.2T_0}$$

Among them, the larger the absolute value of the sensitivity, the more significant the impact of this parameter on the model prediction results.

6.2 Solution of the Sensitivity Model

The seven physical quantities for which sensitivity needs to be calculated are: five sub - powers, internal resistance, battery capacity. Temperature and power consumption mode are environmental variables for fixed scenarios and their sensitivities are not calculated.

So a total of nine groups of sensitivities are obtained:

If the absolute value of the sensitivity coefficient > 1, it is highly sensitive; if it is between 0.3 - 1, it is moderately sensitive; if it is < 0.3, it has a weak impact. According to the model in the first question, the specific reference power values depend on the power consumption mode:

Low - power consumption: Screen power consumption is 0.1W, CPU power consumption is 0.05W, and network power consumption is 0.2506W.

Medium - power consumption: Screen power consumption is 0.3W, CPU power consumption is 0.775W, and network power consumption is 0.272W.

High - power consumption: Screen power consumption is 0.5W, CPU power consumption is 1.5W, and network power consumption is 0.39W. The reference value of static power consumption is 0.065W for all, and GPS power consumption is 0.12W.

Table 5: Calculation Table of Sensitivity Coefficient

Environment/Parameters	SOC	$P_{network}$	P_{GPS}	P_{screen}	P_{static}	P_{CPU}	R_0
$P_{low};T_{low}$	1	-0.44	-0.211	-0.175	-0.114	-0.088	-0.019
$P_{low};T_{mid}$	1	-0.434	-0.208	-0.173	-0.112	-0.087	-0.006
$P_{low};T_{high}$	1	-0.433	-0.207	-0.173	-0.112	-0.086	-0.003
$P_{mid};T_{low}$	1	-0.198	-0.087	-0.218	-0.047	-0.565	-0.081
$P_{mid};T_{mid}$	1	-0.185	-0.082	-0.204	-0.044	-0.528	-0.018
$P_{high};T_{low}$	1	-0.231	-0.071	-0.297	-0.039	-0.891	-0.385
$P_{high};T_{mid}$	1	-0.165	-0.051	-0.211	-0.027	-0.636	-0.039
$P_{high};T_{high}$	1	-0.161	-0.049	-0.206	-0.027	-0.621	-0.019

In conclusion, the most important factors affecting the mobile phone's usage time are battery capacity and CPU power consumption. Temperature mainly affects battery life by changing the internal resistance rather than directly altering the power consumption structure.

6.3 Model Assumption Revision and Result Improvement Analysis

Hypothesis Change 1: Modify the previous assumption that the battery capacity remains constant.

In the previous research, we always assumed that the effective battery capacity remained constant at 4.5Ah. However, in the real working environment of lithium - ion batteries, its value is closely related to other physical quantities of the lithium - ion battery. According to the Peukert effect, the working state of a lithium - ion battery can be described within a certain range:

$$t = \frac{C}{I^k}$$

Where t is the discharge time, C is the nominal capacity of the battery, I is the discharge current, and k is the Peukert exponent (the specific value is determined by the battery type, and for lithium - ion batteries, it is usually taken as 1.05).

Therefore, when I increases, due to limiting factors such as the polarization effect, the effective battery capacity will decrease, no longer conforming to the model assumptions in this paper previously.[8]

Meanwhile, temperature significantly impacts battery capacity. Lithium - ion batteries' discharge behavior varies under low - and high - temperature conditions. At low temperatures, internal diffusion slows, reducing effective capacity. At high temperatures, per the van 't Hoff rule, the reaction rate rises, increasing the apparent available capacity, yet accelerating battery aging to some degree. The lower the temperature, the higher the capacity loss rate.

$$C_{eff}(T) = C_{ref} \cdot e^{-\alpha(T_{ref}-T)}$$

The above formula is a modified common empirical formula, where α is the temperature - sensitive coefficient. This formula fits the experimental data better in the low - temperature region of the battery's operation.

Combining the Peukert effect and the temperature effect, the following formula can be formed to correct the battery capacity:

$$C_{eff}(I,T) = C_{ref} \cdot I^{-(k-1)} \cdot e^{-\alpha(T_{ref}-T)}$$

Where k is the Peukert exponent and α is the temperature - sensitive coefficient, which is generally taken as 0.005.

Introducing the correction term can reduce the simulation deviation of the battery working time. In the high - power scenario of the second question, the simulated time is approximately one - fifth of the measured value of the high - load dataset of the University of Maryland. After correction, with a power of 8W, the simulated usage time is 1.3h, which is highly close to the measured value, verifying the effectiveness and rationality of the correction term.

Hypothesis Change 2: Modify the previous assumption that the power remains constant.

In actual application scenarios, the mobile phone will automatically trigger a protection mechanism when the battery level is low. Also, influenced by user behavior, first, users can manually switch to power - saving mode, which leads to functions like automatic screen brightness adjustment. Second, in our previous research, we assumed that the user always maintains only one power - consumption mode when using the phone. However, in real life, people usually use various power - consumption modes, which overthrows our previous assumption of constant power. The power expression is corrected by making the power change with the State of Charge (SOC). The lower the SOC, the lower the power.

$$P_{cpu}(SOC) = P_{cpu,0}(1 - k_1(1 - SOC))$$

This will make the final time prediction higher than the simulation duration of the original model. So, although it is theoretically reasonable, it is not conducive to fitting real - world data.

Hypothesis Change 3: Modify the previous assumption that temperature only affects internal resistance.

Apart from the temperature - induced impact on battery capacity mentioned above, temperature can also cause changes in the OCV (Open - Circuit Voltage) curve and polarization time constant, which may lead to the premature shutdown of the mobile phone.

The ranking of key factors affecting the mobile phone's usage duration (although demonstrated previously, the results were not presented intuitively): battery capacity - CPU power consumption (extremely sensitive under high load) - network power consumption (obvious in low - power situations) - screen power consumption - GPS power consumption - static power consumption.

Conditional factors: The internal resistance itself has little impact, but it is very important under specific conditions of low temperature and high power.

In conclusion, the physical model proposed in this study has the advantages of a controllable response to changes in key input parameters and insensitivity to disturbances in secondary parameters. It maintains a consistent prediction trend under different physical assumptions and reduces the error compared with real data, indicating that the model has good stability, anti - disturbance ability, and structural robustness. Therefore, this model can not only be used for predicting battery life under the current given working conditions but also has potential applicability for generalization to power - consumption scenarios of similar mobile devices.

7 Multi - dimensional Optimization Strategies for Extending the Battery Life of Smartphones

7.1 Establishment and Analysis of User Behavior Model Based on Factor Changes

The power consumption of a mobile phone battery consists of multiple components such as screen

brightness and processor load. Therefore, we first establish a model for the proportion of power consumption of each behavior.

Based on the relationship between current and power- $I=\frac{P}{V}$, when the voltage is constant, the ratio of power is equal to the ratio of current. So, when the mobile phone battery is completely depleted, the proportion of power consumption of each behavior is:

$$W_x=\frac{\int_0^{T_{empty}} P_x(t)dt}{\int_0^{T_{empty}} P(t)dt} \tag{14}$$

Among them, T_{empty} is the time required for the battery to be depleted, $P_x(t)$ is the power of a certain behavior- x, and $P(t)$ is the total power of the mobile phone ($P(t)=\sum_k P_k(t)$).

Under the average equivalent current, there are $T_{empty}\approx\frac{SOC_0\cdot C}{\overline{I}}$. Therefore, when the power corresponding to a certain factor $\Delta\overline{P}$ drops by $\Delta\overline{I}$, it will inevitably cause the current to drop by \(c\) accordingly. At this time, the new current becomes $\overline{I}'=\overline{I}-\Delta\overline{I}$, and then:

$$\frac{\Delta T}{T}=\frac{T'-T}{T}=\frac{\overline{I}}{\overline{I}-\Delta\overline{I}}-1=\frac{\Delta\overline{I}}{\overline{I}-\Delta\overline{I}} \tag{15}$$

T is the original time required for the battery to be depleted, and ΔT is the time required for the battery to be depleted after the current change. Suppose the contribution of factor ΔT to the total power consumption is reduced by r proportionally compared with the original, then the corresponding current reduction is:

$$\Delta\overline{I}=r\cdot W_x\cdot\overline{I} \tag{16}$$

Substitute into Equation (15) to get:

$$\frac{\Delta T}{T}=\frac{r\cdot W_x}{1-r\cdot W_x} \tag{17}$$

Advice:

The above - mentioned battery life change model indicates that users don't need to know the specific value of W_x. They can compare the power consumption of different activities through the mobile phone's battery power distribution. When the contribution of each factor x to the total power consumption is reduced proportionally by r compared with the original, the larger the value of W_x, the greater the impact of this factor x on the battery depletion time. Reducing the power consumption of factor x can significantly improve the battery's endurance.

Since users use mobile phones for different purposes and in different ways, the proportion of power consumption of factor x in each user's mobile phone varies. Therefore, different suggestions are put forward below for different users to extend the battery life.

Table 6: Targeted suggestions for different users to extend battery life

Different usage modes of mobile phones	The power that is mainly affected	Suggestions
Long - term high - brightness screen	P_{screen}	Reduce the screen brightness and shorten the screen - on time
Playing games or live - streaming for a long time	P_{CPU}	Reduce the time of high - load activities of the processor
Poor network connection and multiple network re - connections	$P_{network}$	Switch to a network with high stability and strong signal
The mobile phone is in positioning mode	P_{GPS}	Turn off location services

In order to more intuitively demonstrate the improvement of the battery life after the improvement, the impact of changing different factors on the battery life is drawn.

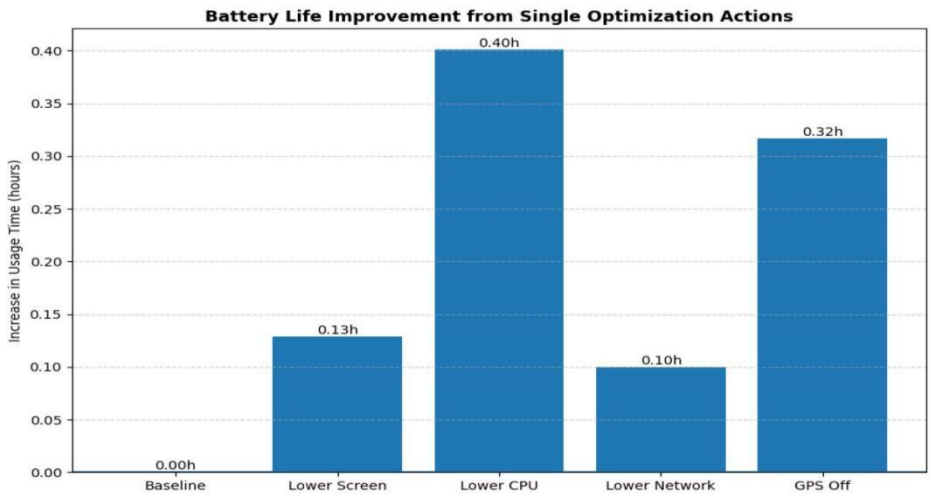

Figure 10: Battery Life Improvement from Single Optimization Actions

With other variables kept constant, reducing the screen brightness, processor load, and network connection by 10% respectively, and turning off the GPS, we can observe that the battery life is extended to varying degrees. Among these, the processor load has the greatest impact on the battery life, while the network connection has the least impact, which is consistent with real - life situations. This further proves the reliability of the model.

7.2 OS Power - Saving Strategy Model Based on Target Optimization

During the process of users using mobile phones, the mobile phones need to meet users' usage requirements. At the same time, the power consumption of the mobile phones should be minimized to extend the battery life. Therefore, a multi - objective optimization model for mobile phone power consumption and users' usage requirements is established.

The current of a mobile phone changes with the variation of its load. The total current can be regarded as the sum of the basic current to maintain the operation of the mobile phone and the additional current caused by different user operations, that is：

$$I(t) = I_{base}(t) + I_{cul}(u(t), t) \tag{18}$$

Where $u(t)$ represents the operating conditions of the mobile phone, including factors affecting the

mobile phone current such as screen brightness and CPU load, $I_{base}(t)$ is the basic current that cannot be changed by the user, and $I_{cul}(u(t),t)$ is the current that can be changed by the user through adjusting the brightness, etc.

During the process of users using the mobile phone, the demand value provided by the mobile phone to the user should not be less than the minimum value of the user's demand, that is:

$$Q(u(t)) \geq Q_{\min} \tag{19}$$

Where $Q_{\min}$ is the minimum value of the usage requirements that the mobile phone needs to meet for the user, and $Q(u(t))$ is the demand value provided by the mobile phone to the user at time t.

During the use of the mobile phone, the temperature will also have a corresponding impact, so it is required that $temperature(t) \leq temperature_{\max}$.

In order to meet the user's requirement for a long battery life of the mobile phone, that is $\max(T_{empty})$, it can be transformed into minimizing the power consumption of the mobile phone per unit time, that is:

$$\min \int_{t_1}^{t_2} I(t,u(t))dt \tag{20}$$

Since users expect both a long battery life and the mobile phone to meet their needs as much as possible, a bi - objective optimization model is established based on this:

$$y = \int_{t_1}^{t_2} a \cdot I(t,u(t)) - b \cdot Q(t,u(t))dt \tag{21}$$

In summary, the bi - objective optimization model is:

$$\min \int_{t_1}^{t_2} a \cdot I(t,u(t)) - b \cdot Q(t,u(t))dt \tag{22}$$

The constraint conditions are: $\begin{cases} Q(u(t)) \geq Q_{\min} \\ temperature(t) \leq temperature_{\max}(t) \end{cases}$.

Advice:

For the operating system: intelligently adjust the brightness, automatically adjust the network, and automatically control whether to turn on or off the GPS according to the actual situation. This way, when the user doesn't need a certain function, it can remain off, reducing battery drainage and extending the battery life. When the user needs it, the phone can automatically change its state to meet the user's needs and enhance the user experience.

7.3 Battery Aging Model

In real life, with continuous charging and discharging of the battery, battery aging and a decrease in battery life are inevitable. Therefore, it is necessary to establish a battery aging model.

The battery capacity decreases with the number of charging cycles. To this end, the established battery aging model is:

$$C(N) = C_0 - k \cdot \ln(1+N)$$

Among them, $C(N)$ represents the number of charging times, N represents the battery capacity

after $C(N)$ charging times, k represents the aging coefficient, with a value of 0.02 here.

Advice:

As the number of charging times rises, battery aging rate exhibits a logarithmic change: fast initially, then slowing, and finally stabilizing.

For users, increasing charging times cause continuous battery capacity decline and aging. Thus, they should regularly check battery health and replace the battery to extend its life.

For the operating system, it can monitor charging and stop when the battery reaches a certain level to prevent overcharging - induced capacity damage. Also, it can cut unnecessary power consumption by managing background processes.

Promotion feasibility

1. Similar chemical properties: All portable devices use lithium - ion batteries, and their battery capacities are affected by factors such as the number of charging cycles and temperature, which is consistent with the model established for smartphones.

2. Identical state variables: Portable devices (tablets, laptops) all use SOC as an important indicator of the remaining battery power, and the change of SOC also follows the law of charge conservation, the same as that of mobile phones.

3. Identical optimization goals: Portable devices aim to maximize the battery life while meeting the basic needs of users, which is consistent with the dual - objective model of "maximizing battery life" and "meeting user needs" established for smartphones.

4. Identical factors affecting power consumption: Both portable devices and smartphones are affected by factors such as screen usage, processor load, network connection, and global positioning system.

8 Model Evaluation and Further Discussion

8.1 Strengths

1. Based on Thevenin equivalent circuit, according to the law of charge conservation and Arrhenius law, the temperature correction of the internal resistance of the mobile phone battery is carried out, which enhances the interpretability of the model. At the same time, the continuous-time dynamics model is used to predict the state of charge (SOC), and the data is verified with the public data set, resulting in a small error.

2. By combining Monte Carlo simulation for local and global sensitivity analysis, the impact of parameter changes on the battery depletion time is clarified, providing a basis for solving the battery endurance problem and realizing the process from theory to application.

3. The battery endurance optimization model has a dual - goal of "maximizing the endurance time" and "meeting user needs". It can extend the endurance time while meeting user needs, making it more in line with reality and the model more reasonable.

8.2 Weaknesses

1. This paper assumes that battery endurance is independent of the discharge rate and does not take into account the interaction between battery aging and the discharge rate.

2. The working scenarios are limited, only considering the temperature range from 0°C to 40°C, without taking into account low - temperature and high - temperature environments.

8.3 Further Discussion

1. In the mobile phone battery life optimization model, regarding the aging coefficient, factors such as the depth of charge and the charging rate are considered for their impacts. Optimization is carried out through specific experimental data, Bayesian optimization and other methods to make it more in line with the actual situations in real life.

2. Based on users' mobile phone usage habits, prediction of battery depletion is conducted. The power consumption of the mobile phone is adjusted through the Markov chain, thereby enhancing the mobile phone's battery life.

9 Conclusion

Based on physical laws, this paper builds a battery life model using the battery's physical mechanism and measured data. The model accounts for various power consumptions and chemical properties, accurately predicting the impact of factor changes on battery life.

To understand the battery's internal discharge, a physical model is constructed. After data verification with a public dataset, Monte Carlo simulation quantifies uncertainty. Uncertainty analysis shows the model is robust and can handle parameter fluctuations.

Global and local sensitivity analyses conclude that the most crucial factor for battery life is electrical capacity, followed by CPU power consumption, which helps extend battery life.

The model can predict battery depletion time and clarify the influence degree of different factors. It offers practical advice to smartphone users, a solid scientific basis, and has broad application prospects in tech services.

Thanks to its modular structure, the study's framework can be extended to other electronics or electrochemical systems, with vast potential for future expansion in intelligent battery optimization and management.

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Report on Use of AI

GPT-5-Thinking

Query 1: The authors consulted an AI tool regarding the standard formulation and applicability of Thevenin’s theorem, including the physical interpretation of the equivalent voltage source and internal resistance in circuit modeling.

Output:

The AI tool provided the general statement of Thevenin’s theorem, indicating that any linear electrical network can be equivalently represented by an ideal voltage source in series with an internal resistance at its terminals, and clarified that the theorem applies under linear and steady-state or quasi－steady-state conditions. This response was used solely to confirm the theoretical basis of the equivalent circuit representation.

Query 2: The authors asked the AI tool to clarify the standard definition of electrical power and its physical meaning in terms of voltage and current.

Output:

The AI tool provided the general expression P=UIP = UIP=UI and explained that electrical power represents the rate of energy transfer per unit time. This explanation was used only to confirm the physical interpretation of power within the model framework, rather than for specific calculations

Query 3: For the sensitivity analysis, the authors consulted the AI tool regarding the conventional definition and interpretation of local sensitivity coefficients.

Output:

The AI tool explained that local sensitivity coefficients are commonly defined as normalized partial derivatives of the model output with respect to input parameters, and are used to quantify the relative influence of small parameter perturbations. This information was used only to clarify the meaning of the sensitivity metric.

Query 4: To support the temperature-dependent modeling of battery internal resistance, the authors asked the AI tool about the standard mathematical form and physical meaning of the Arrhenius law.

Output:

The AI tool described the Arrhenius law as an empirical relationship commonly used to characterize the exponential dependence of reaction rates or equivalent parameters on temperature, and provided its general form to clarify the theoretical origin of the temperature correction term.